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EXCERPT OF DISSERTATION

Synthesis of Butenolides via Transition Metal-Catalyzed Cyclization Reaction of 2,3-Allenoic Acids*

MA Sheng-Ming† YU Zhan-Qian

(*State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, Shanghai* 200032, *China*)

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Abstract Butenolides are a class of compounds of current interest due to their occurrence in natural products and the potential broad range of biological activities of natural and unnatural products. During the course of our study on transition metal-catalyzed cyclization reaction of 2,3-allenoic acids, we have developed palladium(Ⅱ)-catalyzed reaction of 2,3-allenoic acids with (ω-1)-alkenyl halides to introduce sp^3-hybridized carbon atoms at the β position of butenolides. In this reaction, we observed a new Pd(Ⅱ)-catalytic cycle which combines carbopalladation with repeated dehydropalladation/hydropalladation-β-dehalopalladation. Furthermore, a new area of oxidative cyclization-dimerization reaction of two functionalized allenes has been established. In this area, we have developed oxidative dimeric cyclization reaction of 2,3-allenoic acids to afford bi-butenolides. Moreover, oxidative cyclization-dimerization reaction of 2,3-allenoic acids with 1,2-allenyl ketones was also established to furnish interesting compounds with two different cycles in a single step.

Key words butenolides, 2,3-allenoic acids, palladium, cyclization, oxidative

CLC O627

Butenolides, structural unit in many natural products, are important intermediates in organic synthesis[1]. Butenolide-containing compounds are considered as potential insecticides, bactericides, fungicides, antibiotics, anticancer agents, anti-inflammatories, allergy inhibitors, antisoriasis agents, cyclooxygenase inhibitors, and phospholipase A_2 inhibitors, etc[2]. Thus, much attention has been paid to the development of new methods for the synthesis of these interesting compounds[3].

Transition metal-catalyzed methodology has been proven to be one of the most powerful pathways for the formation of carbon-carbon bonds as well as carbon-heteroatom bonds[4]. Based on the development of synthetic

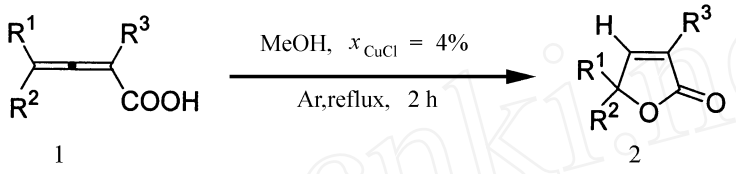
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† E-mail: masm@mail.sioc.ac.cn

methodologies of butenolides in our group[5], we have developed transition metal-promoted or catalyzed cyclization reactions of 2,3-allenoic acids/salts for the synthesis of several kinds of substituted butenolides.

Under the catalysis of CuCl, the β-unsubstituted butenolides were formed smoothly from 2,3-allenoic acids (Table 1)[6].

Table 1　CuCl-catalyzed cyclization of 2,3-alkadienoic acids



1　　2

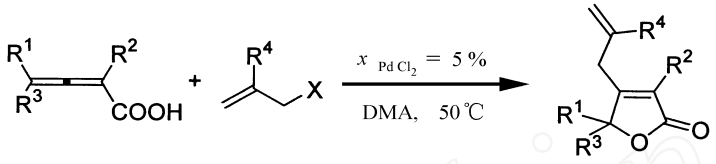
Entry	1			Product 2	Yield of 2/(%)
	R^1	R^2	R^3		
1	Ph	H	*n*-C_3H_7 (1a)	2a	97
2	Ph	H	CH_3 (1b)	2b	95
3	Ph	H	$PhCH_2$ (1c)	2c	90
4	Ph	H	allyl (1d)	2d	96
5	α-Naphthyl	H	CH_3 (1e)	2e	92
6	α-Naphthyl	H	*n*-C_3H_7 (1f)	2f	96
7	α-Naphthyl	H	allyl (1g)	2g	92
8	CH_3	H	*n*-C_3H_7 (1h)	2h	78
9	CH_3	H	$PhCH_2$ (1i)	2i	90
10	*n*-C_3H_7	H	CH_3 (1j)	2j	87
11	*n*-C_3H_7	H	$PhCH_2$ (1k)	2k	85
12	*n*-C_3H_7	H	allyl (1l)	2l	91
13	*n*-C_6H_{13}	H	CH_3 (1m)	2m	85
14	Ph	C_2H_5	CH_3 (1n)	2n	93
15	*n*-C_7H_{15}	H	H (1o)	2o	81

This reaction is fairly general and the yields range from good to excellent. Compared to known methodologies ($AgNO_3$ or H^+ was used as catalyst), the current catalyst (CuCl) has the advantage of being relatively cheap, more abundant, and low molecular weight (compared to $AgNO_3$), and high yielding (at least comparable to $AgNO_3$ and much higher than that of the H^+-catalyzed reaction), which makes this methodology more attractive.

On the other hand, others in our group have applied the palladium(0)-catalyzed cyclizations of 2,3-allenoic acids or salts for the synthesis of butenolides. In those reactions, we could easily introduce an aryl/vinyl substituent into the β-position of the butenolides[5]. Nevertheless, under those established reaction conditions the sp^3-hybridized carbon atom could not be introduced into the product via the direct reaction of the allenoic acids and sp^3-carbon halides due to the intrinsic stability problem of alkyl palladium species. Although we have successfully developed an alternative two-steps method for synthesis of the β-alkyl substituted butenolides[7], we wish to develop a new method with which the sp^3-hybridized carbon atom could be directly introduced into butenolides.

Under the catalysis of palladium dichloride, β-sp^3-hybridized carbon atom substituted butenolides could be synthesized from the easily available allylic halides and 2,3-allenoic acids (Table 2)[8].

Table 2 The Coupling Reaction of 2,3-Allenoic Acids and Allylic Halides



1 3 4

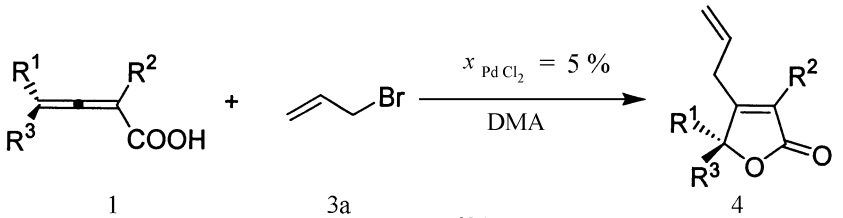
Entry	1			3		Time/h	4	Yield /(%)
	R^1	R^2	R^3	R^4	X			
1	Ph	*n*-C_3H_7	H (1a)	H	Br (3a)	17	4a	88
2	Ph	CH_3	H (1b)	H	Br (3a)	17.5	4b	83
3	1-Naphthyl	*n*-C_3H_7	H (1c)	H	Br (3a)	21	4c	78
4	1-Naphthyl	CH_3	H (1d)	H	Br (3a)	16	4d	81
5	Ph	$PhCH_2$	H (1e)	H	Br (3a)	16	4e	71
6	1-Naphthyl	*n*-C_3H_7	H (1c)	Ph	Br (3b)	24	4f	93
7	Ph	*n*-C_3H_7	H (1a)	CH_3	Cl (3c)	24	4g	89
8	1-Naphthyl	*n*-C_3H_7	H (1c)	CH_3	Cl (3c)	15.5	4h	95
9	Ph	*n*-C_3H_7	H (1a)	Br	Br (3d)	21	4i	77
10	1-Naphthyl	*n*-C_3H_7	H (1c)	Br	Br (3d)	23	4j	79
11	1-Naphthyl	CH_3	H (1d)	Br	Br (3d)	23.5	4k	69
12	Ph	Allyl	H (1f)	H	Br (3a)	17	4l	72
13	1-Naphthyl	CH_3	H (1d)	CH_3	Cl (3c)	24	4m	76
14	CH_3	$PhCH_2$	H (1g)	H	Br (3a)	17	4n	52
15	CH_3	$PhCH_2$	H (1g)	H	Br (3a)	17.5	4n	73[b]
16	Ph	CH_3	Et (1h)	H	Br (3a)	19	4o	83
17	4-Br-C_6H_5	*n*-C_3H_7	H (1i)	H	Br(3a)	11.5	4p	75

From Table 2, it should be pointed out that the yields range from moderate to excellent. Allylic chloride is equally active. The substituents of the 2,3-allenoic acids may be aryl, allyl, benzyl or alkyl group. As a result of introducing an allylic group into the products, further elaboration of the C═C bond would provide an efficient access to other butenolide derivatives.

Under the standard conditions, the chirality transfer is less effective (Scheme 1) probably due to the relatively high reaction temperature. When the reaction was carried out at 25 ℃, the chirality of the starting material was transferred into the product efficiently. Under the new conditions, we easily obtained four optically active butenolides.

Two control experiments were carried out to elucidate the reaction mechanism. The reaction of substrate 1a and 3-chloro-1-butene 3e gave 91 % of the product 4q (Eq. (1)), while the reaction of substrate 1a with 1-chloro-2-butene 3f was relatively slow leading to a 17∶1 mixture of 4q and 4qA in a combined yield of 19 % (Eq. (2)).

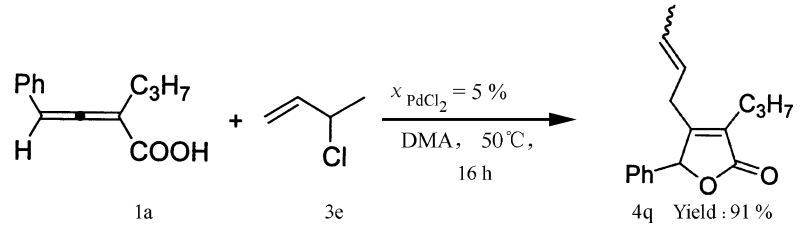
Based on the different results outlined in equations (1) and (2), the reaction may proceed via three consecutive steps: cyclic oxypalladation of the allene, insertion of the C═C bond in allylic halides and β-dehalopalladation.



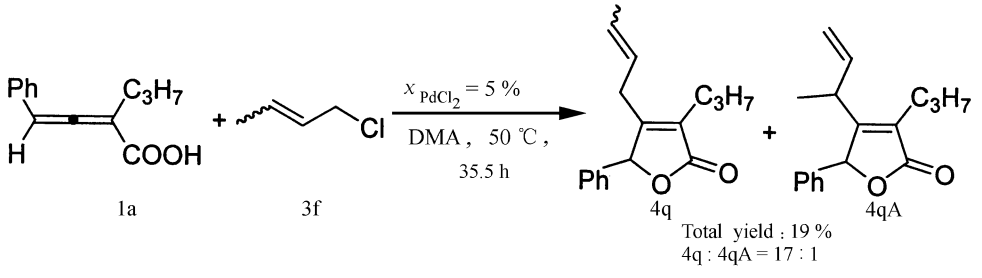
1　3a　4

	T / (℃)	Time / h	
(+)-1a (R^1 = H, R^2 = *n*-C_3H_7, R^3 = Ph) > 98 % ee	50	16	(+)-4a 90 %, 94 % ee
	25	35	(+)-4a 80 %, >99 % ee
(-)-1a (R^1 = Ph, R^2 = *n*-C_3H_7, R^3 = H) >97 % ee	25	36.5	(-)-4a 82 %, 99 % ee
(+)-1d (R^1 = H, R^2 = CH_3, R^3 = Naphthyl) >98 % ee	25	40	(+)-4d 91 %, 90 % ee
	0	84.5	(+)-4d 39 %, 98 % ee
(-)-1d (R^1 = Naphthyl, R^2 = CH_3, R^3 = H) >97 % ee	25	40	(-)-4d 85 %, 92 % ee

Scheme 1　Four optically active products obtained by chirality transfer



1a　3e　4q　Yield: 91 %　(1)

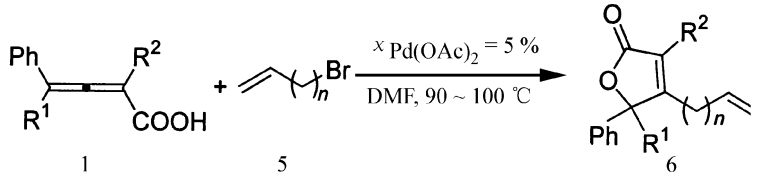


1a　3f　4q　4qA　(2)

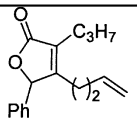
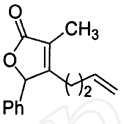
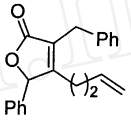
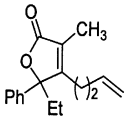
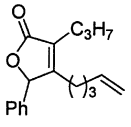
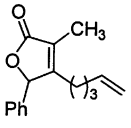
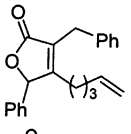
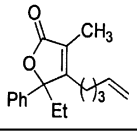
Total yield: 19 %
4q : 4qA = 17 : 1

In order to introduce more sp^3-hybridized carbon atoms between the vinyl group and butenolide, Pd(Ⅱ)-catalyzed coupling cyclyzation reaction of 2,3-allenoic acids with (ω-1)-alkenyl halides was carried out[9]. Some typical results under the established reaction conditions are shown in Table 3. The substitutent of allenoic acids could be aryl, alkyl, and benzyl. 5-Bromo-1-pentene could also be used in this reaction.

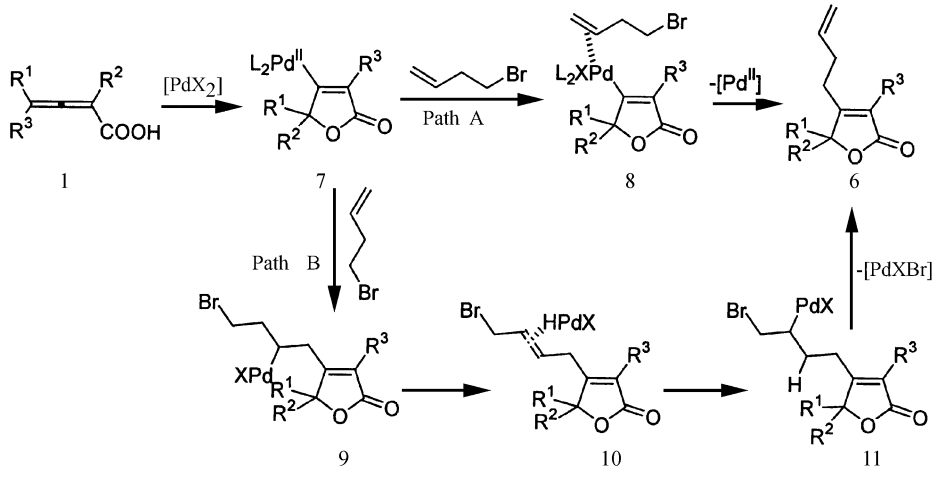
Table 3　The coupling cyclization of 2,3-allenoic acids and (ω-1)-alkenyl halides



1　5　6

Entry	1		5	Time/h	Product 6		Yield/(%)
	R^1	R^2	*n*				
1	H	C_3H_7 (1a)	2 (5a)	1		6a	58
2	H	CH_3 (1b)	2 (5a)	1		6b	43
3	H	$PhCH_2$ (1c)	2 (5a)	1.5		6c	48
4	Et	CH_3 (1d)	2 (5a)	1		6d	36
5	H	C_3H_7 (1a)	3 (5b)	10		6e	45
6	H	CH_3 (1b)	3 (5b)	2		6f	23
7	H	$PhCH_2$ (1c)	3 (5b)	4		6g	30
8	Et	CH_3 (1d)	3 (5b)	1.5		6h	27

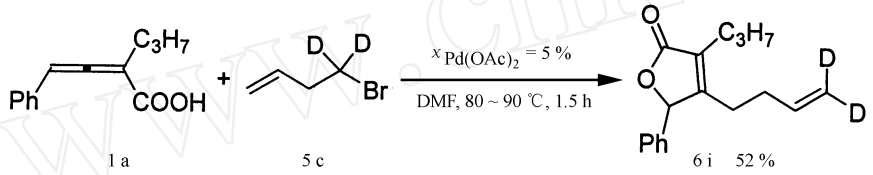
In principle, the presence of the C═C bonds allows the reaction to undergo two possible mechanisms for this transformation (Scheme 2). One is the coordination-directed direct substitution pathway (path A), in which



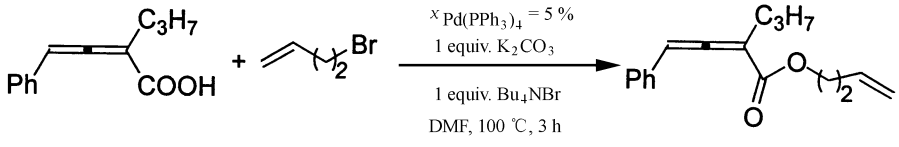
Scheme 2 The possible reaction pathways

$PdCl_2$ coordinates with the double bond remote from the acid group in 2,3-allenoic acids 1. Subsequent cyclic oxypalladation affords palladium intermediate 7, which coordinates with the carbon-carbon double bonds of homoallyl bromide 5a followed by an oxidative addition-reductive elimination process leading to 6. The other alternative pathway is the cyclic oxypalladation-carbopalladation-repeated dehydropalladation/hydropalladation-dehalopalladation pathway (path B).

In order to clarify the mechanism, the reaction of deuterated homoallyl bromide 5c and 4-phenyl-2-propyl-2,3-butadienoic acid 1a was executed. Only product 6i in which the two deuterium atoms located at the terminal carbon atom of the noncyclic carbon-carbon double bonds was obtained in 52 % yield (Eq. (3)). When the reaction was carried out in the presence of $Pd(PPh_3)_4$, only 3'-butenyl 4-phenyl-2-propyl-2,3-butadienoate 12 was obtained in 96 % yield (Eq. (4)).



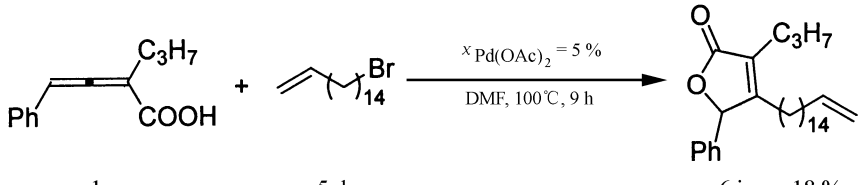
1 a　5 c　6 i　52 %　(3)



1 a　5 a　12　96 %　(4)

According to the experiment facts, the reaction may proceed through a cyclic oxypalladation-carbopalladation-repeated dehydropalladation-hydropalladation-β-dehalopalladation process. To the best of our knowledge, this is the first example of a Pd (Ⅱ)-catalyzed cycle that combines carbopalladation-repeated dehydropa-lladation/hydropalladation-β-dehalopalladation. Based on these results, we are interested to see how far a repeated dehydropalladation and hydropalladation sequence can go.

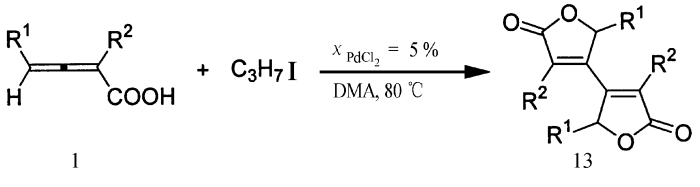
We found that the reaction of 4-phenyl-2-propyl-2,3-butadienoic acid 1a and 16-bromo-hexadec-1-ene 5d also gave the corresponding product 6j in 18 % yield (Eq. (5)). In this reaction the palladium could go over fourteen carbon atoms!



1 a　5 d　6 j　18 %　(5)

Furthermore, it would be interesting to synthesize butenolides from the dimerization of two allenes. In this area, three oxidative systems were established for converting 2,3-allenoic acids to the non-readily available bi-butenolides (Table 4)[10].

Table 4　Oxidative dimeric cyclizing-coupling reaction of 2,3-allenoic acids

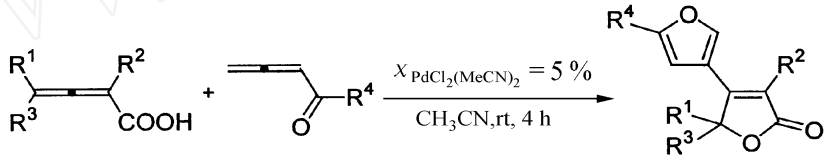


1　13

Entry	Substrate 1		Time/h	Product 13	Yield/(%)	R*S*/R*R*
	R^1	R^2				
1	Ph	*n*-C_3H_7 (1a)	10	13a	74	1/1.87
2	Ph	CH_3 (1b)	4	13b	64	1/1.62
3	Ph	$PhCH_2$ (1c)	11.5	13c	75	1/1.53
4	α-Naphthyl	CH_3 (1d)	4	13d	71	1/1.28
5	α-Naphthyl	*n*-C_3H_7 (1e)	4.5	13e	64	1/1.43
6	CH_3	$PhCH_2$ (1f)	3.5	13f	72	1.04/1

Furthermore, the oxidative cyclization-dimerization reaction between two classes of allenes with different functionalities was also realized. The reaction of 2,3-allenoic acids and 1,2-allenyl ketones provide an efficient route to polysubstituted 4-(3'-furanyl)-2(5H)-furanones (Table 5)[11], which are not readily available from the known methods.

Table 5 Pd(Ⅱ)-Catalyzed Cyclizative Cross Coupling Reaction of 2,3-Allenoic Acids and 1,2-Alkenyl Ketones



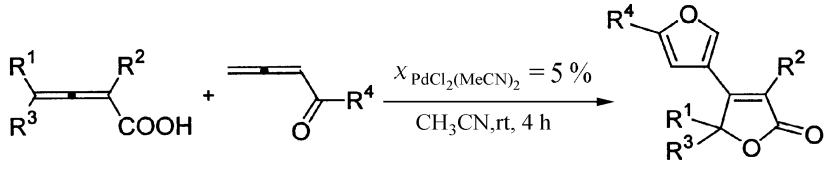
1 14 15

Entry	1			14	15	Yield/(%)
	R^1	R^2	R^3	R^4		
1	Ph	*n*-C_3H_7	H (1a)	CH_3 (14a)	15a	90
2	Ph	CH_3	H (1b)	CH_3 (14a)	15b	73
3	Ph	$PhCH_2$	H (1c)	CH_3 (14a)	15c	92
4	Ph	Allyl	H (1d)	CH_3 (14a)	15d	86
5	α-naphthyl	CH_3	H (1e)	CH_3 (14a)	15e	84
6	α-naphthyl	*n*-C_3H_7	H (1f)	CH_3 (14a)	15f	86
7	CH_3	$PhCH_2$	H (1g)	CH_3 (14a)	15g	90
8	Ph	CH_3	Et (1h)	CH_3 (14a)	15h	85
9	*n*-C_7H_{15}	H	H (1i)	CH_3 (14a)	15i	69
10	4-Br-Ph	*n*-C_3H_7	H (1j)	CH_3 (14a)	15j	70
11	Ph	*n*-C_3H_7	H (1a)	*n*-C_4H_9 (14b)	15k	84
12	Ph	CH_3	H (1b)	*n*-C_4H_9 (14b)	15l	63
13	Ph	Allyl	H (1d)	*n*-C_4H_9 (14b)	15m	66
14	α-naphthyl	CH_3	H (1e)	*n*-C_4H_9 (14b)	15n	79
15	Ph	CH_3	Et (1h)	*n*-C_4H_9 (14b)	15o	86
16	*n*-C_7H_{15}	H	H (1i)	*n*-C_4H_9 (14b)	15p	69
17	Ph	*n*-C_3H_7	H (1a)	$PhCH_2$ (14c)	15q	74
18	Ph	CH_3	H (1b)	$PhCH_2$ (14c)	15r	86
19	Ph	Allyl	H (1d)	$PhCH_2$ (14c)	15s	82
20	α-naphthyl	CH_3	H (1e)	$PhCH_2$ (14c)	15t	90
21	Ph	CH_3	Et (1h)	$PhCH_2$ (14c)	15u	84
22	*n*-C_7H_{15}	H	H (1i)	$PhCH_2$ (14c)	15v	71

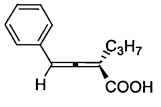
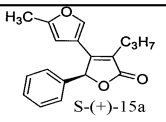
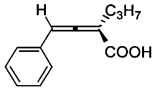
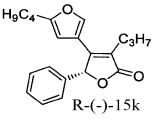
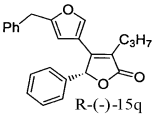
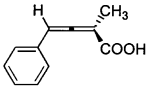
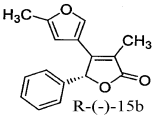
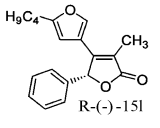
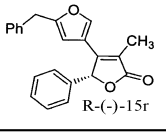
This reaction can be furnished under mild conditions (rt, 4 h) and the yields are from moderate to fairly good. It should be noted that this is the first example of oxidative sequential cyclizative cross coupling reaction of two differently functionalized allenes to give interesting compounds with two different cycles in a single step. In this reaction, differently substituted 2,3-allenoic acids could be used: monosubstituted 2,3-allenoic acids reacted with the 1,2-allenyl ketones smoothly to give the corresponding products; disubstituted 2,3-allenoic acids are more popularly used in this reaction. The substituents of 2,3-allenoic acids could be aryl, alkyl, benzyl and allyl. With the fully substituted 2,3-allenoic acids, the reaction also afforded the products in fairly good yields. Moreover, the substituent of the 1,2-allenyl ketones could be the alkyl or benzyl group.

Under the standard reaction condition, the efficiency of chirality transfer is very high. The highly optically active butenolides could be easily formed from the optically active 2,3-allenoic acids (Table 6). The absolute configuration of the product was determined by the X-ray single crystal diffractional analysis. From (*S*)-2,3-allenoic acids, the product obtained is in *S* configuration (Scheme 3).

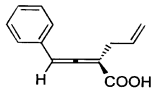
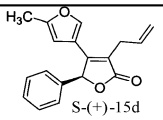
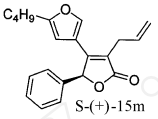
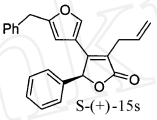
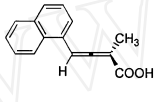
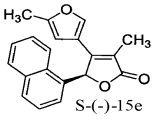
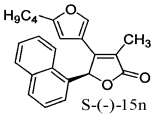
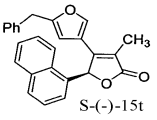
Table 6　Pd(Ⅱ)-Catalyzed Cyclizative Cross Coupling Reaction of 1,2-Allkenyl Ketones with Optically Active 2,3-Allenoic Acids

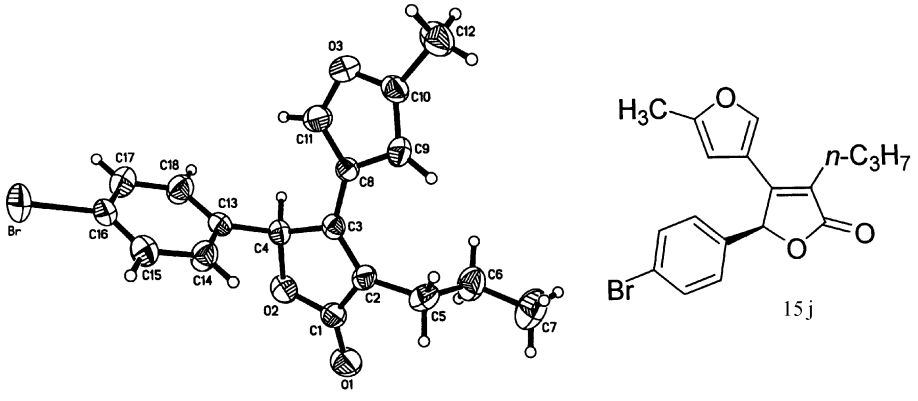


1　　14　　15

Entry	1	14	15	Yield of 15/(%)	Ee of 15/(%)
1	 S-(+)-1a (98 % ee)	14a	 S-(+)-15a	87	95
2	 R-(-)-1a (97 % ee)	14b	 R-(-)-15k	75	> 99
3	R-(-)-1a (97 % ee)	14c	 R-(-)-15q	86	> 99
4	 R-(-)-1b (99 % ee)	14a	 R-(-)-15b	76	97
5	R-(-)-1b (99 % ee)	14b	 R-(-)-15l	81	99
6	R-(-)-1b (99 % ee)	14c	 R-(-)-15r	72	96

(续表)

Entry	1	14	15	Yield of 15/(%)	Ee of 15/(%)
7	 S-(-)-1d (98 % ee)	14a	 S-(+)-15d	73	97
8	S-(-)-1d (98 % ee)	14b	 S-(+)-15m	67	99
9	S-(-)-1d(98 % ee)	14c	 S-(+)-15s	67	98
10	 S-(+)-1e (98 % ee)	14a	 S-(-)-15e	74	97
11	S-(+)-1e (98 % ee)	14b	 S-(-)-15n	76	99
12	S-(+)-1e (98 % ee)	14c	 S-(-)-15t	87	> 99

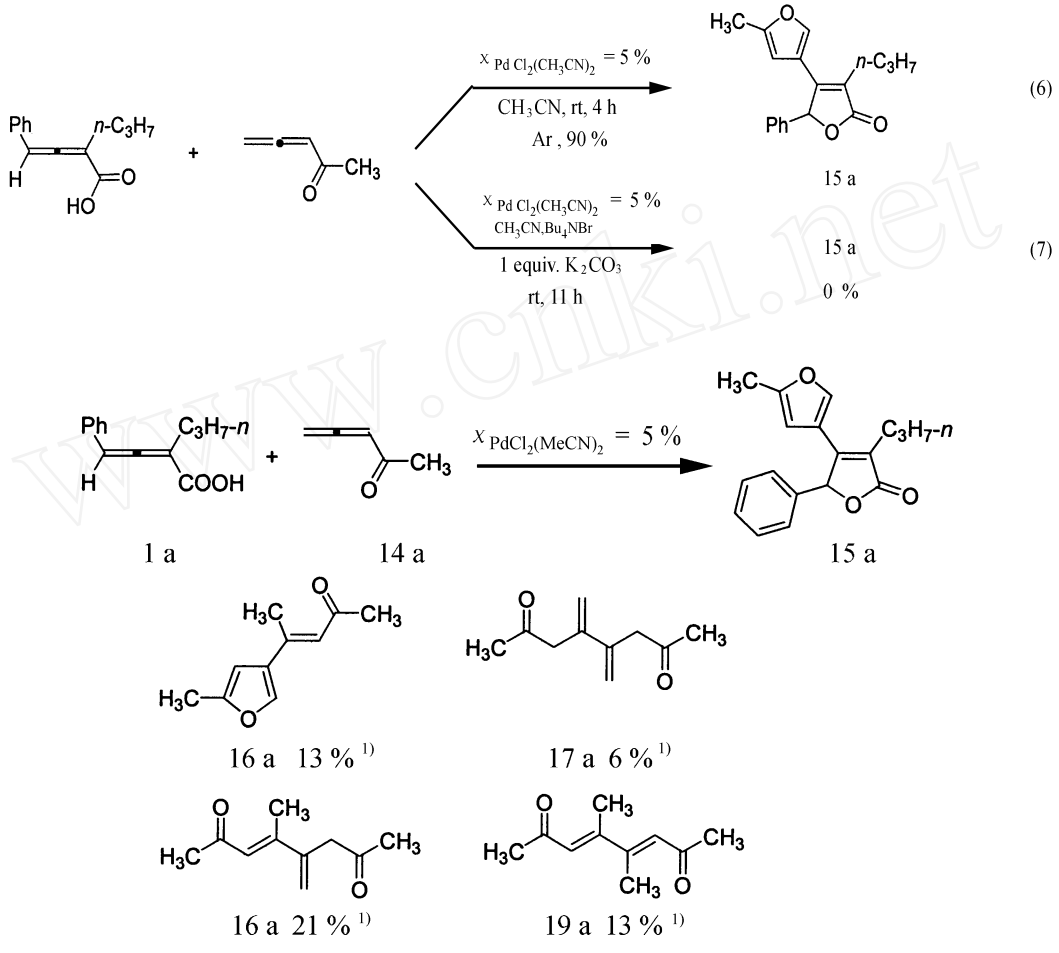


Scheme 3 The X-ray structure analysis shows that compound 15j is in S configuration

Comparing the products with starting materials, we observed that two hydrogen atoms were lost after the reaction. Where had they gone? Two control experiments were conducted to answer this question. In an atmosphere of argon, the reaction occurred smoothly to afford 15a in 90 % yield (Eq. (6)), indicating that oxygen does not participate in the catalytic cycle. Another interesting point is that the reaction did not yield 15a in the presence of 1 equiv. of K_2CO_3 (Eq. (7)), indicating that H^+ may be formed from the loss of two protons from both starting materials and plays an important role in this reaction.

Further study shows that the reaction also afforded the homodimerization product of 14a, (i.e., furan 16a), 4,5-dimethyleneocta-2,7-dione 17a and its C═C double bond isomers 18a, and 19a as the byproducts (Scheme

4), indicating that the missing hydrogens were transferred from starting materials into 2,4-dien-1,6-diketones 17a, 18a and 19a.



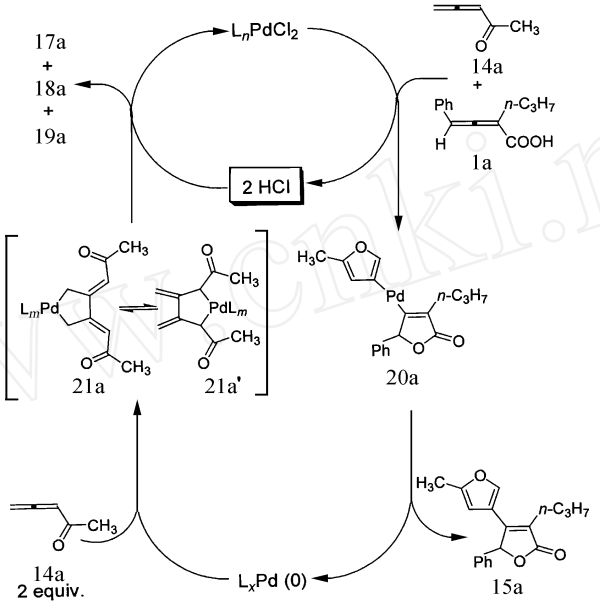
Scheme 4　Oxidative cyclization-dimerization reaction of 2,3-allenoic acid with 1,2-allenyl ketone

Based on these facts, a plausible mechanism was proposed for this reaction (Scheme 5). Firstly, the intermediate 20a is formed from the 1,2-allenoic ketone and 2,3-allenoic acids via double oxypalladation, which is further supported by the highly efficient chirality transfer. In our previous work, we observed that the chirality transfer is not efficient in a carbopalladation pathway but very efficient in an oxypalladation pathway[5]. After the reductive elimination of 20a, the product 15a is formed along with the Pd(0) species. Then, the Pd(0) species cyclometalate with 2 equiv. of penta-3,4-dien-2-one to form the cyclic Pd intermediate 21a and/or 21a', which may be protonated with the *in-situ* generated HCl (2 equivalents) to form 17a, 18a and 19a and regenerate L_nPdCl_2 to furnish the catalytic cycle.

In conclusion, we have developed CuCl-catalyzed cyclization reaction to synthesize β-unsubstituted butenolides. Palladium(Ⅱ)-catalyzed reactions of 2,3-allenoic acids with (ω-1)-alkenyl halides were found to be able to introduce sp^3-hybridized carbon atoms to the β position of butenolides. Among those reactions, we observed a new Pd(Ⅱ)-catalytic cycle which combines carbopalladation with repeated dehydropalladation/hydropalladation-β dehalopalladation. Furthermore, a new area of oxidative cyclization-dimerization reaction of two functionalized

1) Yields are calculated based on penta-3,4-dienone used

allenes has been established. In this area, we have developed oxidative dimeric cyclization reaction of 2,3-allenoic acids to afford bi-butenolides. Moreover, oxidative cyclyzation-dimerization reaction of 2,3-allenoic acids and 1,2-allenyl ketones was established to furnish interesting compounds with two different cycles in a single step.



Scheme 5 Proposed catalytic cycle for the oxidative cyclization-dimerization reaction

Acknowledgemet Shanghai Municipal Committee of Science and Technology are greatly appreciated.

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过渡金属催化的 2,3-联烯酸环化合成 γ-丁烯酸内酯

麻生明　俞瞻前

(中国科学院上海有机化学研究所金属有机化学国家重点实验室, 上海　200032)

摘　要　γ-丁烯酸内酯是一类具有重要潜在活性的化合物. 发展了二价钯催化的 2,3-联烯酸和(ω-1)-烯基卤化物的偶联关环反应. 反应经过了二价钯催化的氧钯化-碳钯化,重复的去钯氢化/钯氢化及 β 脱卤钯化反应. 同时还发展了两种联烯之间的二聚环化反应,两分子 2,3-联烯酸的反应生成二丁烯酸内酯,2,3-联烯酸与 1,2-联烯酮反应则生成呋喃基取代的 γ-丁烯酸内酯.

关键词　丁烯酸内酯,2,3-联烯酸,钯,环化,氧化

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俞瞻前:博士研究生在读期间,荣获 2003 年度中国科学院院长奖学金特别奖.

导师麻生明:主要从事有机合成、金属有机等领域的研究.